

International Islamic University Chittagong
Department of Computer Science and Engineering
B. Sc. in CSE Midterm Examination, Autumn 2025
Course Code: CSE 4747

Course Title: Mathematical Analysis for Computer Science
Total marks: 30 Time: 1 hour 30 minutes

[Answer all the questions. Figures in the right-hand margin indicate full marks.]

		M	CO	DL
1.				
a)	source peg destination peg 1 2 3 A B C initial state The Tower of Hanoi is a classic recursive problem. You are given n disks and three pegs (A, B, C). The objective is to move all disks from peg A to peg C following these rules: ○ Only one disk can be moved at a time. ○ A disk can only be placed on top of a larger disk or an empty peg. ○ All disks start on peg A in decreasing order of size. Let $T(n)$ denote the minimum number of moves required to solve the Tower of Hanoi problem with n disks. i) Write the recurrence relation for $T(n)$. ii) Solve the recurrence to find a closed-form expression for $T(n)$. iii) Find the value of $T(7)$.	6	CO1	C2, C3
b)	Use the Akra-Bazzi formula to find the asymptotic bounds for the following recurrence: $T(n) = 2T(n/4) + \sqrt{n}, \quad T(1) = 1 \text{ and } T(n) = \Theta(1)$ Or, Find the solution of the following non-homogeneous recurrence equation: $a_n = 2a_{n-1} - a_{n-2} + 2^n \text{ with } a_0 = 1 \text{ and } a_1 = 2$	4	CO1	C3
2.				
a)	Evaluate the following expression: $\sum_{k=50}^{75} k^2$	2	CO1	C2, C3, C4

b)	Compute J(53) using both the recurrent and binary properties of the Josephus problem while eliminating every 2 nd person.	2	C01	C3
c)	Find a closed form for the following expression: i) $n! = \prod_{i=1}^n i$ ii) $\sum_{k=1}^n (2k+1)^3$ Or, The maximum overhang in the book staking problem is given by the following recurrence: $B_1 = \frac{1}{2}$ $B_{n+1} = B_n + \frac{1}{2(n+1)}$ Now convert this recurrence into sum problem and then find a solution of the sum.	5	C01	C3
d)	Write down a faster alternative of the following code segment i.e. optimize it: <pre>int N=100; float sum = 0; for (int i = 1; i <= N; i++) sum += i*i*i;</pre>	1	C02	C3
3.				
a)	You're preparing a candy gift bag with four types of candies: Chocolates, Wafers, Gums, and Mints. ❖ The number of Mints must be even. ❖ The number of Chocolates must be a multiple of 5. ❖ There can be at most four Gums. ❖ There can be at most one Wafers. Let nnn be the total number of candies in the bag. Determine the number of ways to prepare the candy gift bag.	5	C02	C2, C3, C4, C5
b)	Write a Generating Function for the Fibonacci series.	2	C02	C2
c)	Enumerate the scaling and addition operations on Generating function with example.	3	C02	C2, C3,

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International Islamic University Chittagong

Department of Computer Science & Engineering

Mid Term Exam, Autumn 2025

Course Code: CSE-4741 Course Title: Computer Graphics

Time: 1.5 Hours

Total Marks: 30

Answer all the questions. Figures in the right margin indicate full marks.
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- | | CO | Marks |
|---|-----|-------|
| 1a) Define computer graphics. Explain how a real-world 3D object mapped to display/pixel coordinate using graphics pipeline. | CO1 | 4 |
| 1b) If we want to resize a 1024×768 image to one that is 640 pixels wide with the same aspect ratio, what would be the height of the resized image. How much memory will be needed to store the resized image, if we use 3 bits per primary color in direct coding for each pixel. | CO2 | 4 |
| 1c) Justify the reason, why an additional black pigment is needed in printer. | CO1 | 2 |
| 2a) Determine first two quadrant points of a circle with radius 10 using Bresenham's circle algorithm and symmetry method. | CO3 | 5 |
| OR | | |
| 2a) Write the steps that are required to scan convert a circle using Midpoint circle algorithm. Evaluate the equation $p1=1-r$ from midpoint circle algorithm. | | |
| 2b) Apply boundary fill algorithm to fill a region with seed point (5, 5), if the boundary color is white. Mention the steps to fill that region. | CO2 | 3 |
| OR | | |
| 2b) Determine which raster location would be chosen by Bresenham's algorithm when scan converting a line from pixel coordinate (3, 3) to (9,8). | | |
| 2c) How image representation will be jagged in pixel coordinate by aliasing? Propose some methods to solve this problem? | CO1 | 2 |
| 3a) Find the new point after reflection of a point (4, 3) about a line $y = 6$ using composite transformation. | CO1 | 4 |
| 3b) Suppose a line with points (5,6) and (10, 7) is rotated 45 degrees in counter clock-wise direction about the point (5,6). Find the new coordinate of line. | CO1 | 4 |
| 3c) Prove that, Inverse geometric transformation is the coordinate transformation with example. | CO3 | 2 |
| OR | | |
| 3c) Find the steps to mirror reflect of an object about a line L. | | |

International Islamic University Chittagong

Department of Computer Science & Engineering

Mid Term Examination Autumn 2025

CSE 4745 Numerical Methods

Total Marks: 30 Time: 90 Minutes

[Answer *all* the *three* questions. Figures in the right-hand margin indicate full marks.]

- 1.a) Draw the *block diagram* of the process of numerical computing. 1.5 CO1
C1
- b) Write short notes on: 2 CO1
i) Roundoff errors ii) Truncation errors C1
- c) Describe the *Banker's rounding rule* with examples. Use *banker's rounding rule* to round off the following numbers to four significant figures - 4 CO1
i) 505.5371 ii) 0.859458 iii) 0.0001425 iv) 3.14285714 C2
OR
How to count significant digits of a number? Use *Banker's rounding rule* to round off the following numbers to four significant figures -
i) 505.5371 ii) 0.859458 iii) 0.0001425 iv) 3.14285714
- d) What do you mean by *absolute error* and *relative error*? If $X = 0.0650958$ is rounded off to four significant figures compute the absolute error and relative error in X . 2.5 CO1
C2
- 2.a) What is *Horner's rule*? Evaluate the polynomial $f(x) = 4x^5 - 6x^4 + 2x^2 - 23x + 15$ using Horner's rule at $x = M$, where M denotes the *last digit* of your ID number. 2 CO2
C3
- b) Find the root of the equation $x^3 - 7x + 4 = 0$, correct to two decimal places, by using the *false position method*. [False position formula: $x_0 = x_1 - (f(x_1)(x_2 - x_1)) / (f(x_2) - f(x_1))$] 5 CO2
C3
OR
Find the root of the equation $x^3 + 5x - 11 = 0$, correct to two decimal places, by using the *secant method*. [secant formula: $x_{n+1} = x_n - ((f(x_n) * (x_n - x_{n-1})) / (f(x_n) - f(x_{n-1})))$]
- c) Write an algorithm to determine *all possible real roots* of an equation. 3 CO1
C2
- 3.a) Find the missing term in the following data using *Newton's Binomial Expansion formula*. 2 CO2
C3
x 0 1 2 3 4
y 3 9 24 - 61
- b) Derive the *Newton's backward interpolation formula*. 4 CO1
C2
OR
Derive the *Newton's divided difference formula*.
- c) The following data gives the melting point of an alloy of lead and zinc, where t is the temperature in degree Celsius ($^{\circ}C$) and P is the percentage of lead in the alloy. Using a *suitable interpolation formula*, find the melting point of the alloy containing 84 percent [if the last digit of your ID is *even*] / 53 percent [if the last digit of your ID is *odd*], with the help of the following data. 4 CO2
C3

P	40	50	60	70	80	90
t	170	204	236	260	288	332

International Islamic University Chittagong
Center for General Education (CGED)

Midterm Examination: Autumn 2025

Course Code: URIH-4701

Time: 1 hours and 30 minutes.

Program: Undergraduate

Course Title: A Survey of Islamic History & Culture

Full Marks: 30

Instructions:

- i. All Questions are Compulsory.
- ii. Figures in the right margin indicate full marks.
- iii. Course Learning Outcome (CLO) and Bloom's levels are mentioned in additional columns.

Bloom's Levels of the Questions.

Letter of Symbol	R	U	App	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

	Text of the Questions	Marks	Bloom's Level	CLO
1	Elucidate the concept of <i>khilafah</i> in Islam. What are the qualities and responsibilities of a <i>khalifah</i> ?	10	U	CLO1
2	Who was <i>Abu Bakr (R.)</i> ? point out the significant contributions of <i>Khalifah Abu Bakr (R.)</i> as the savior of Islam.	10	E	CLO3
3	Explain the administrative development under <i>Khalifah Umar (R.)</i> . How is modern administration indebted to these developments? Or Assess the contributions of <i>Khalifah Othman</i> , and <i>Ali (R)</i> to the development of <i>Khilafah</i> and World Civilization.	10	An	CLO3

International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE Midterm Examination, Autumn 2025

Course Code: CSE 4877

Course Title: Machine Learning and Data Mining

Total marks: 30 Time: 1 hour 30 minutes

[Figures in the right-hand margin indicate full marks.]

1. Mar CO
- a) Discuss whether or not each of the following activities is a data mining task: ks 2 1
- (a) Dividing the customers of a company according to their gender.
 - (b) Sorting a student database based on student identification numbers.
 - (c) Predicting the outcomes of tossing a (fair) pair of dice.
 - (d) Monitoring the heart rate of a patient for abnormalities.
- b) List any two major steps involved in the KDD process. 2 1
- c) Consider the following customer dataset: 1+ 2
- 2+3

Table 01:

Customer ID	Age	Gender	Income(\$)	Join Date
C001	25	Male	45000	2023-05-12
C002	32	Female	52000	2022-11-07
C003	29	Male	(Missing)	2023-01-20
C004	41	Female	61000	(Missing)
C005	23	Male	38000	2023-08-05

Answer the following questions:

- i) Identify the type of each attribute. Which attributes are categorical and which are numerical?
- ii) Impute the missing values in Income and Join Date.
- iii) Normalize the attributes Age and Income from the given dataset using appropriate normalization methods. Justify your choice of method for each attribute.

2. 2 2
- a) Given the following dataset of values: 2 2
- 48, 66, 52, 5, 43, 65, 49, 62, 52, 54, 53
- Perform outlier detection using the Boxplot (Interquartile Range – IQR) method.
- b) A sample Data Table containing attributes of mixed type. 3 2

Table 02:

Object Identifier	test-1 (nominal)	test-2 (ordinal)
1	code A	Excellent
2	code B	Fair
3	code C	Good
4	code A	Excellent

Using the dataset above, construct the dissimilarity matrices for both the nominal and ordinal attributes.

- c) Consider the following two documents:

Text 1:

Data preprocessing is a process where data is collected, cleaned, and transformed for useful insights. Effective data handling improves algorithm performance. Several algorithm techniques help in data processing and pattern recognition. Machine learning algorithms can automatically find patterns in data, enabling intelligent learning from examples.

Text 2:

In modern data science, algorithm development plays a vital role. Each algorithm is tested on different data samples to train the model. Through learning and pattern discovery, the analysis of results becomes more accurate. Good data preparation ensures better predictions and efficient output detection.

Assume there are 10 unique words in the vocabulary:

[data, mining, algorithm, statistics, learning, model, clustering, pattern, AI, analysis]

Compute the cosine similarity between the two document vectors and interpret the result.

3.

- a) What type of sampling strategy is applied in the following scenarios? Provide a short justification for your answer.

4 2

i) A researcher plans to study customer satisfaction in a busy supermarket. Since interviewing every customer is not practical, the researcher decides to collect data from every 10th customer who enters the store.

ii) A university wants to conduct a survey on the satisfaction level of its students regarding online learning. The university has five faculties — Engineering, Business, Science, Arts, and Law. The survey team decides to select 50 students from each faculty, ensuring that every faculty is equally represented in the sample.

OR

Suppose a group of 12 sales price records has been sorted as follows:

5,10,11,13,15,35,50,55,72,92,204,215. Partition them into three bins by each of the following methods:

- (a) equal-frequency (equal-depth) partitioning
- (b) equal-width partitioning
- (c) smoothing by bin means

b) Consider the following transaction database:

Table 03:

Transaction ID	Items Bought
T1	A, B, D, E
T2	B, C, D
T3	A, B, C, E
T4	A, B, C, D, E
T5	B, C, E

1+2 3
+3

Answer the following questions:

- With minimum support = 3, find all frequent 1-itemsets (L1).
- Using the frequent 1-itemsets (L1), generate all candidate 2-itemsets (C2) and compute their support counts. Then identify the frequent 2-itemsets (L2) for minimum support = 3.
- Suppose, you found that
 $\text{sup}(\{B, C\}) = 3$
 $\text{sup}(\{B\}) = 3$

Calculate the confidence for the rule $B \rightarrow C$. Based on your result, what marketing or shelf-placement strategy could the shop use to increase the sales of C?

OR

1+2 3
+3

Answer the following questions for the transaction database (Table 3) given in question 3(b):

- Reorder the transactions in descending order of frequency for FP-Growth.
- Construct the FP-Tree from the reordered transactions. Indicate item counts in each node.
- From the FP-Tree, derive the conditional pattern bases for the frequent items B and E and interpret the results.